E5: free checks on P-192 and P-224 (no ECDLP run)

Status. E5a is measured under a protocol frozen before the run (`e5/PROTOCOL.md`, commit `ff44dd462`). E5b is a **model**: constants from dry run 2 at 24 bits (scratch, `PROTOCOL.md` of E1 not frozen), projected by derived formulas. **No ECDLP run. No security claim. E5b is not a feasibility claim.** All statements below are labelled measured, derived or model.

The owner's question. On P-192 and P-224 the generator `G` is fixed by the standard and every key is `[k]G`. Does the fixed `G` give index calculus anything? E5 runs the only checks that are feasible on the real curves:

- **E5a:** do the first `2^24` multiples of the standard `G` look different, in their abscissae, from random points?
- **E5b:** what would a base built from `G` (known logs, no linear algebra) cost at 192 and 224 bits, against the abscissa baseline, BSGS and rho?

Answer in one line. E5a: no. None of 20 pre-registered tests rejects after Holm at 0.05; the smallest p is 0.22. E5b (model): the known-log base is a generic precomputation table. At the plan's `B = ⌈r^{1/3}⌉` it sits `2^32` (P-192) and `2^37` (P-224) from rho at one key. At its own optimal size it comes within `2^0.4` to `2^0.8` of rho at one key, using a table of `2^95.5` or `2^111.5` entries. It never reaches rho with shared distinguished points (Kuhn–Struik), and it never reaches BSGS at equal memory.

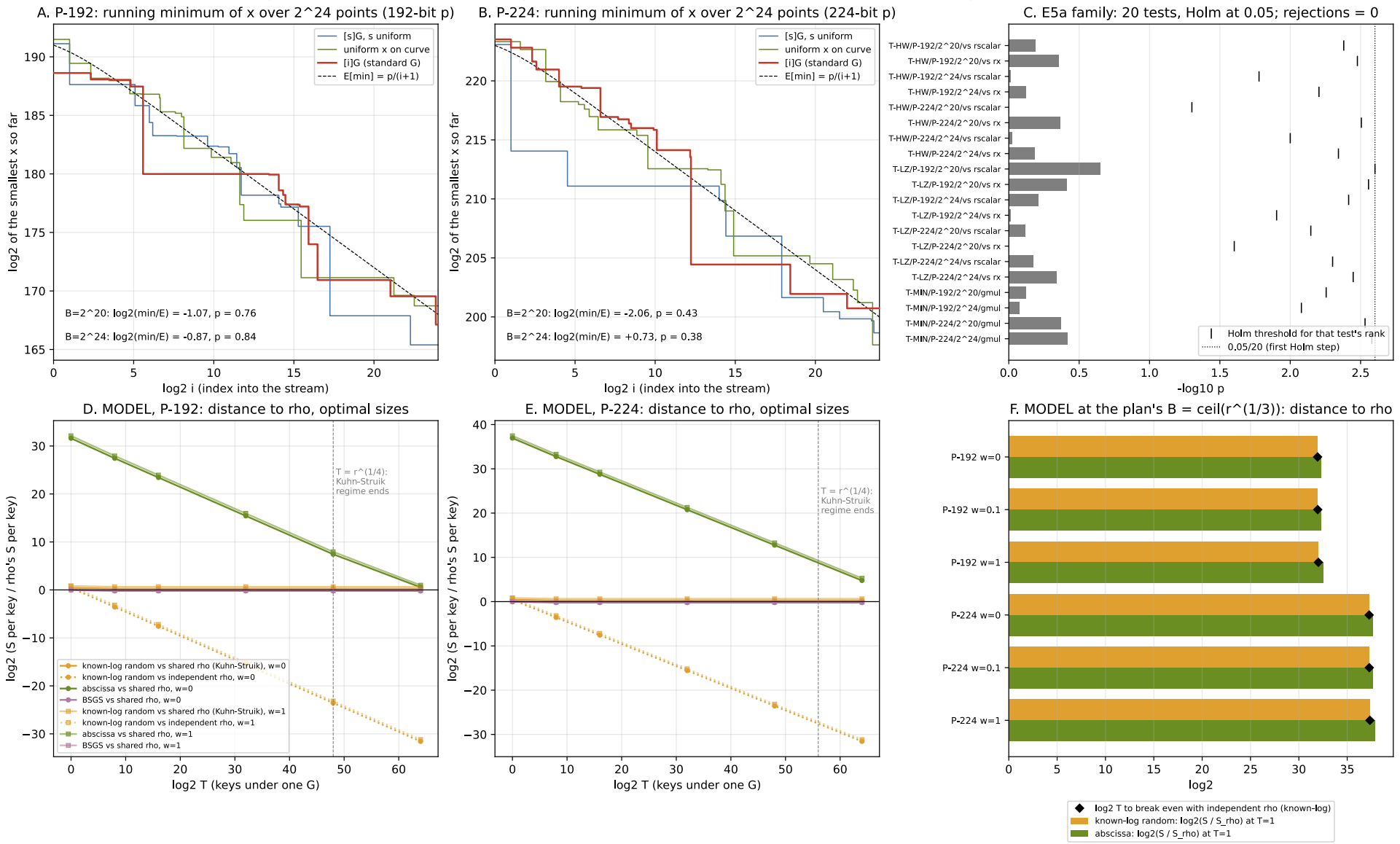


Figure 1: E5a x-statistics (A–C, measured) and E5b cost model (D–F, model)

Figure 1. A, B (measured): running minimum of x over the first 2^24 elements of each stream, against the order-statistic expectation `p/(i + 1)`. C (measured): the 20 tests of the family, `-log10 p` against each test's Holm threshold. D, E (model): log2 of S per key over rho's S per key against the number of keys `T`, each arm at its optimal size, at `w = 0` (solid markers) and `w = 1` (faded). F (model): the plan's equal `B`, distance to rho at `T = 1` and break-even `T` against independent rho. E5a seed 20261019; E5b constants from sessions ECBS1h0331ce29276e and ECBS1hd98aace0ae34. Source: `e5/plot_e5.py` (lays out the JSON only).

E5a: the first 2^24 multiples of G (measured)

What ran. `cargo run --release --example generator_effect_e5a` (source) on `CurveParams::p192()` and `CurveParams::p224()`. It builds three streams of `2^24` canonical abscissae per curve: `gmul = x([i]G)`, `rscalar = x([s]G)` with `s` uniform in `[1, n−1]`, and `rx`, uniform `x ∈ [0, p)` on the curve. Seeds and chunking are as in the protocol. `B = 2^20` is the prefix of each stream. Arithmetic: 4-limb Montgomery field (`ct_bignum::Uint<4>`), Jacobian points, batched inversion per `2^16` points, and an 8-bit fixed-base comb for `[s]G`. Output: `e5/e5a.json`. Log: `e5/e5a.log`. Run report: `reports/run-20261019T2159Z/`.

Checks (all pass). `G` is on each curve and `[n]G = O`. 202 BigUint cross-checks per curve: `x([i]G)` at `i ∈ {1, 2, 3, 1000, 2^16, 2^16+1, 2^20, 2^24}` plus the first 64 of `gmul`, 64 `rscalar` points through `Point::scalar_mul_vartime`, and 64 `rx` right-hand sides by Euler's criterion. 0 mismatches. The planted positive control (rx with element `B/2` replaced by the first valid abscissa, `x = 0` on P-192 and `x = 3` on P-224) is detected in all 4 cells: p = 0, 0, 2.3e-61 and 3.7e-60 against 0.0025. A second run from the final committed source gave identical JSON except for wall times.

The family (measured). Holm–Bonferroni over 20 tests, family-wise α = 0.05. **Rejections: 0.** The prediction holds.

curve	B	T-LZ vs rscalar	T-LZ vs rx	T-HW vs rscalar	T-HW vs rx	T-MIN (gmul)
P-192	2^20	0.225	0.389	0.646	0.439	0.759
P-192	2^24	0.613	0.969	0.971	0.759	0.843
P-224	2^20	0.763	0.991	0.992	0.432	0.427
P-224	2^24	0.672	0.458	0.948	0.655	0.382

p-values. T-LZ: two-sample chi-square on leading-zero bins `0…K−1, ≥K`, `K = log2 B − 6`, df = K. T-HW: two-sample chi-square on Hamming-weight bins `ℓ/2 ± 3σ` with pooled tails, df = 44 (P-192) and 48 (P-224). T-MIN: two-sided, `F(u) = 1 − e^{−u}`, `u = B·x_min/p`. The smallest p is 0.225 against a first Holm threshold of 0.0025.

The statistics themselves (measured).

curve	B	set	mean HW	smallest x: log2	its index	log2(x_min / (p/(B+1)))
P-192	2^24	gmul	96.0002	167.13	15 193 886	−0.87
P-192	2^24	rscalar	96.0012	165.39	5 076 147	−2.61
P-192	2^24	rx	95.9987	168.73	5 991 468	+0.73
P-224	2^24	gmul	111.9963	200.73	4 133 403	+0.73
P-224	2^24	rscalar	111.9994	198.65	13 191 441	−1.35
P-224	2^24	rx	111.9992	197.63	12 482 318	−2.37
P-192	2^20	gmul	95.9999	170.93	91 246	−1.07
P-224	2^20	gmul	111.9954	201.94	354 421	−2.06

The full histograms, all record lows and the `2^20` controls are in `e5a.json`. Bit length is `ℓ − lz`, so it carries the same information as the leading-zero test. The shortest abscissa among the `gmul` points is 168 bits on P-192 and 201 bits on P-224.

Calibration, outside the family (measured). Each of the 12 (curve, B, set) cells was also tested against its analytic law: leading zeros of a uniform `x ∈ [0, p)`, and Hamming weight Binomial(ℓ, 1/2). The smallest of the 24 p-values is 0.022 (P-192, `2^20`, gmul, leading zeros). Under any multiplicity correction over 24 tests that is not significant. Every T-MIN on the controls lies in `[0.11, 0.85]`.

What E5a rules out (derived from the measured minima). T-MIN would have rejected for `log2(x_min/E) ≤ −9.6` at the first Holm step. Directly, without any statistics: no `±[i]G` with `1 ≤ i ≤ 2^24` has an abscissa below `2^167.1` on P-192 or below `2^200.7` on P-224. In particular the lexicographically first point (`x = 0`, `x = 3`) is not `±[i]G` for any such `i`.

Power, stated plainly (derived). The two-sample chi-square tests at `2^24` per sample detect a change of about 1% in a bin holding 1/64 of the mass, and much less in the large bins. A generator chosen to hide structure outside these statistics would pass. E5a is a test for visible structure, not a proof of its absence.

Cross-reference: trapdoor program slot S4

`nist_trapdoor_slot_program_20261011/PROTOCOL.md` §10 lists two pending S4 items: E4a, rigid-derivation tests against four candidate rules, and E4b, a protocol audit.

- **E4a, rule "`G = [h]·(first point)` with `h = 1`":** E5a **narrows it, in one direction only.** It excludes `P_first = ±[i]G` for `1 ≤ i ≤ 2^24`, that is `G = [h]·P_first` with `h ≡ ±i⁻¹ (mod n)`. That family includes `h = 1`, which closes the rule as written for both curves (the census already had it from `G.x ≠ x_first`). It does **not** cover small `h > 1`, because `h⁻¹ mod n` is then a large

scalar. The direct check, `[h]·P_first` for `h ≤ 2^24` compared with `G`, is a cheap follow-up and is not run here. (Run since as E5c, `E5c.md`: no match for `h ≤ 2^24`, both roots, both curves.)

- **E4a, SHA-1 rules** (`x = SHA-1(seed ‖ i)`, the X9.62 construction): not touched.
- **E4b** (protocol audit): not touched.
- **P-192 specifically.** `nist_seed_provenance_20261016/README.md` attributes P-192's `G` to Lenstra's FreeLIP `zrandom` (seeds 20000–20517). That is random output, and E5a's null result is what that provenance predicts. For P-224 the provenance is unexplained; E5a finds no visible structure in its first `2^24` multiples.

S4 stays pending. E5a closes the `h = 1` case, which was already closed, and adds the inverse family `P_first = ±[i]G`, `i ≤ 2^24`. It closes no other listed item.

E5b: the known-log base at 192 and 224 bits (MODEL)

What it is. `cargo run --release --example generator_effect_e5b` (source) reads the 384 measured (non-warm-up) records of dry run 2's two phase sessions and `summary2.json`'s yield block. It extracts the constants below and evaluates derived cost formulas at `r = n(P-192) ≈ 2^192` and `r = n(P-224) ≈ 2^224`. Output: `e5/e5b.json`. **The dry run reports no fitted exponents** (two curves at one size); the exponents here come from the structure of each algorithm, not from a fit. That is the main caveat of this section.

Constants (measured at 24 bits, pooled over both curves).

constant	value	source
GAE per scalar multiplication, per bit of r	1.373	known-log random, factor-base phase (B charged scalar mults)
per-key walk overhead	34.4 scalar mults	relations phase GAE minus walk steps
verify	1.00 scalar mult	verify phase
additions per trial	0.959	walk steps / trials
unpriced ops per trial (lookup + guard probe)	2.00	relations phase
abscissa: relations per base element `f_R`	0.554	relations / B
abscissa: LA row ops per (B · relations)	0.053	linear algebra phase
abscissa: base native ops per element	3.06	Legendre + square roots
hit rate, measured / `2B(B+1)/r`	0.973	3000-target yield block
BSGS giant steps, measured / `r/(2(2M+1))`	1.033	bsgs-neg-B
rho-neg S at w = 0 / 0.1 / 1	1.044 / 1.137 / 1.975	rho-neg, canonicalisations at w

Formulas (derived). `E = B(B+1)` pair-table entries; hit rate `h = 0.973 · 2E/r`; `sm = 1.373 · log2 r` GAE; trial `= 0.959 + 2w` GAE.

- known-log random: `B·sm + E(1+w) + T·[(34.4 + 1.0)·sm + trial/h]`.
- abscissa: `w·3.06·B + E(1+w) + (f_R·B − 1)·trial/h + w·0.053·B·f_R·B + T·[(35.4)·sm + trial/h]`.
- `bsgs.negation_table`, M baby classes: `sm + M(1+w) + T·[sm + 1.033·(1+w)·r/(2(2M+1))]`.
- rho: `s_rho(w)·√r` per key, keys independent (dry run's convention). A second variant shares distinguished points across keys (Kuhn–Struik 2001): total `≈ (2/√π)·√T·s_rho(w)·√r`. This is an asymptotic valid for `T ≪ r^{1/4}` (`2^48`, `2^56`); points beyond are drawn but flagged `ks_regime_ok = false` in the JSON.
- Pricing: the dry run left the IC arms' pair-table inserts uncounted while it priced BSGS's inserts at `w`. Here both are priced at `w`, symmetrically.

At the plan's equal B = ⌈r^{1/3}⌉ (model)

curve	log2 B	log2 table	online probes per key, log2: known-log / BSGS same memory / abscissa first key	log2(S/S_rho) at T=1, w = 0 / 0.1 / 1: known-log	same: abscissa	break-even T vs independent rho (log2), known-log	vs shared rho
P-192	64.0	128.0	63.0 / 62.0 / 126.2	31.9 / 32.0 / 32.0	32.3 / 32.3 / 32.5	31.9 / 32.0 / 32.0	never
P-224	74.7	149.3	73.7 / 72.7 / 147.5	37.3 / 37.3 / 37.4	37.6 / 37.7 / 37.9	37.3 / 37.3 / 37.4	never

At this `B` the pair table (`2^128`, `2^149`) is the whole cost. The known-log arm breaks even with *independent* rho only after about `2^32` (P-192) or `2^37` (P-224) keys. With shared distinguished points, rho's per-key cost falls as `1/√T` and the fixed table never catches it. Against BSGS holding the same table, the known-log arm pays 1.91 / 2.10 / 2.94 times the online cost per key at `w = 0 / 0.1 / 1`: random sums cover `2E` residues per probe by chance, where BSGS strides cover `2(2M+1)` deterministically. It never catches BSGS at equal memory.

Each arm at its own optimal size (model)

curve	w	known-log: log2 B, log2 table at T=1	log2(S/S_rho) at T=1	log2(S/S_shared rho), T ≥ 2^8	break-even T vs independent rho	BSGS optimum vs shared rho (log2)	abscissa at its optimum, log2(S/S_rho) at T=1
P-192	0	47.7, 95.5	0.43	0.25	2^0.86 ≈ 1.8	−0.04 (T=1), −0.21	31.6
P-192	0.1	47.8, 95.6	0.51	0.34	2^1.02 ≈ 2.0	−0.02, −0.20	31.7
P-192	1	47.9, 95.8	0.82	0.65	2^1.64 ≈ 3.1	+0.04, −0.13	32.1
P-224	0	55.7, 111.5	0.43	0.25	2^0.86 ≈ 1.8	−0.04, −0.21	36.9
P-224	0.1	55.8, 111.6	0.51	0.34	2^1.02 ≈ 2.0	−0.02, −0.20	37.1
P-224	1	55.9, 111.8	0.82	0.65	2^1.64 ≈ 3.1	+0.04, −0.13	37.4

At its optimum (`B ≈ r^{1/4}`, table ≈ `√r`) the known-log arm is plain generic precomputation. It costs about `√(2(1+w)(1+2w)·r/T)` per key, within `2^0.4` to `2^0.8` of rho at one key and below independent rho from two or three keys. That is the Bernstein–Lange shape PLAN §2 named, with memory `2^95.5` (P-192) and `2^111.5` (P-224) entries at one key, growing as `√T`. Against shared rho its per-key cost is a constant factor `2^0.25` to `2^0.65` above, for every `T` in the Kuhn–Struik regime. BSGS at its own optimum ties shared rho to within `2^0.04` at one key and sits `2^0.13` to `2^0.21` below it for more keys; that is inside the model's accuracy (an idealised BSGS against a measured rho overhead). The abscissa baseline's optimum is `B ≈ r^{1/3}` (`2^63`, `2^74`), and it stays `2^32` to `2^37` from rho at one key. Its relation collection is what the known-log base removes.

Distance remaining (model). On the CLAUDE.md criterion, the known-log random-scalar base is an improvement over the abscissa baseline in this model: `2^31` to `2^37` at `T = 1` once each is at its optimum. The reason is that it no longer needs `0.55·B` relations before the first key. It is no improvement over generic algorithms. At equal memory it is 1.9 to 2.9 times BSGS's online cost per key, and it stays `2^0.25` to `2^0.82` above rho with shared distinguished points. None of this is a feasibility statement: the tables hold `2^95` to `2^149` entries.

What the model does not cover. No fitted exponent from a size sweep; 24-bit constants carried to 192 and 224 bits. A per-key walk overhead proportional to `log2 r` (scalar mults). Hit rate from the sumset law, checked only at 24 bits. Memory access cost, which is unpriced. Parallelism. The chained-increment base (B additions instead of B scalar mults), which changes the known-log setup by less than `2^−39` of the table at these sizes. Semaev oracles, for which a set `{[s]G}` has no polynomial description (PLAN §2).

What this says about the owner's question

- **Measured (E5a):** the standard `G`'s first `2^24` multiples carry no visible abscissa structure on P-192 or P-224. Their sizes, bit patterns and smallest values are those of random points.
- **Derived (PLAN §1B) and modelled (E5b):** the only thing a fixed `G` gives index calculus is that logs computed once, base `G`, are reused for every key. A base built from `G` makes those logs free. In the MITM pipeline that base is a generic precomputation table, and generic precomputation is already bounded by rho and BSGS. It does not move an exponent.

Files

file	what
`e5/PROTOCOL.md`	E5a protocol, frozen before the run (commit `ff44dd462`)
`e5/e5a.json`, `e5/e5a.log`	E5a output and log (run 20261019T2159Z)
`e5/e5b.json`	E5b model output
`e5/e5.svg`, `e5/e5.png`, `e5/plot_e5.py`	figure, 150 dpi render, plotting-only source
`E5.typ`, `E5.pdf`, `build_e5_pdf.sh`	this report as PDF
`../../examples/generator_effect_e5a.rs`	E5a scan
`../../examples/generator_effect_e5b.rs`	E5b model
`reports/run-20261019T2159Z/`	run report of the E5a run

Graphs checked and left unchanged: `docs/ic-leaderboard.html` and `docs/ic/LEADERBOARD.md` (no measured IC row; E5b is a model), `docs/index-calculus-scoreboard.html`, and the trapdoor program's figures (S4 is narrowed, not closed). No `benchmarks/` entry, because nothing here measures speed. Conductor: the MCP returned 404 and the CLI has no project configured on this host, so no task was obtained.